

Week 3 — Reading Pack

Curated excerpts from industry white papers, the syllabus readings, and the PMSA items — organized by learning objective

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Required reading before Week 4 · ≈35 minutes (plus the Jin et al. sections, ≈30 minutes)

Assigned alongside: Jin et al., “Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects” (Google, 2017), §§1–3; the PyMC-Marketing MMM example notebook; the MMM Field Guide (reference).

Why a reading pack. MMM vocabulary in pharma is learned from the documents practitioners write and the papers the libraries are built on. Part A sets the stakes in the industry's own words (verbatim, with source, from IQVIA papers in the course library). Part B walks you through the two syllabus readings and what to look for. Part C is pharma promotion at launch. Part D is digital measurement in healthcare — the part of the mix where attribution is closest to being solved and still is not. Part E lists the PMSA items. Read, answer the reflection questions, bring one answer to Week 4. PDFs are in the “reading assignment” folder for this week.

Learning objectives for Week 3 (from the syllabus)

1. Specify a Marketing Mix Model with adstock, saturation, and base/incremental decomposition.
2. Translate a fitted MMM into channel-level ROI and a defensible reallocation recommendation.
3. Recognize identifiability, multicollinearity, and seasonality pitfalls in MMM.

Key concepts: linear and Bayesian MMM; adstock (geometric/Weibull) and Hill saturation; cross-channel attribution vs. MMM vs. multi-touch attribution; calibration with experiments (geo-lift, holdout DMAs).

A. Why the Brand Lead asks the question (objective 2)

Two excerpts set the stakes: commercial leaders say they want to allocate spend with more precision and less delay, and they report returns on analytics investments that — like any ROI claim — should make you ask how they were measured.

Life sciences organizations still need to prepare more effectively for high-value customer interactions, allocate spend with greater precision, understand care pathways and patient journeys, and respond to access barriers and customer behavior. What has changed is the tolerance for delay and ambiguity. Senior decision makers need answers faster, with context they can trust and enough confidence to act.

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

Across the commercial enterprise, the problem is not a lack of imagination. The use cases are clear: sharper field preparation, more responsive omnichannel planning, better spend allocation, improved customer understanding, faster insight generation, and more precise engagement. The constraint is execution. Too many initiatives remain trapped between promising pilots and scaled enterprise value because the operating model has not caught up to the technology agenda.

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

Investment in AI is rising, and the returns are real. Nearly half of the surveyed companies dedicate more than 20% of their commercial budgets to AI, with another quarter investing between 11% and 20%. This level of commitment reflects AI's growing role in commercial strategy with proven results. The returns are

compelling: 58% of leaders report achieving 2X ROI from their AI initiatives within a single year, with 7% seeing 3X or more.

— IQVIA, “AI in Life Sciences Commercialization” (2025 survey of 107 senior commercial leaders), Executive summary

Sales and Marketing: High priority, low adoption — Sales and marketing are rated highest in strategic importance, yet low current average adoption. Despite the promise of AI to drive field productivity, omnichannel targeting, and content personalization, many organizations have not yet industrialized these tools. The barriers are less about ambition and more about scaling enablement across decentralized teams.

— IQVIA, “AI in Life Sciences Commercialization,” AI impact by commercial function

While organizations use varied data sources — such as CRM, field activity, claims, prescriptions, digital engagement, patient support, and market data — these are often siloed and inconsistently structured, limiting their utility for enterprise-scale AI. As one respondent put it, “We have the data — we just can’t use it the way we need to.”

— IQVIA, “AI in Life Sciences Commercialization,” Barriers to scale

Reflection A. “58% report 2× ROI within a year” is a self-reported average return. Using tonight's vocabulary, write the three questions you would ask before believing it: what is the counterfactual (base), is it average or marginal, and what would an experiment look like? Then say which of the data sources in the last excerpt an MMM needs and which it does not.

B. The syllabus readings — what to look for (objectives 1 & 3)

Jin, Wang, Sun, Chan & Koehler, “Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects” (Google, 2017)

This is the paper behind the adstock-and-Hill model you fitted tonight and behind the priors in PyMC-Marketing and Google's Meridian. It is readable; the mathematics is the same as the notebook's. Read §§1–3 and skim the simulation section. Look for:

- **The two transforms.** Carryover is written as a weighted sum of past spend with geometric weights (and a delayed variant whose peak comes after lag 0 — the Weibull-style idea on slide 9). Shape is the Hill function with a half-saturation and a slope parameter — tonight's K and s.
- **Priors on the transform parameters.** The authors put priors on the decay, the delay, and the Hill parameters and sample everything jointly, instead of grid-searching. Note how they justify the choice of prior for each parameter — the homework bonus asks you to do the same.
- **Identifiability, in their words.** They show — with simulated data where the truth is known, exactly as tonight — that different combinations of carryover and shape parameters can fit the observed data almost equally well, and that the estimated ROI can be biased even when the fit is good. Compare with notebook §§5c and §8.
- **ROI and marginal ROI.** They define both from the fitted model (the change in the outcome when a channel's spend is removed or perturbed) and show why the two differ under saturation — slide 11.
- **Their recommendations.** More data (longer series, geo-level rows), informative priors from experiments and past studies, and reporting the uncertainty. The calibration triangle on slide 27 is the practitioner's version.

The paper is public (research.google, “Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects”). A copy is in the reading folder if the instructor was able to add it; otherwise download it from the Google Research site.

PyMC-Marketing — MMM documentation and example notebook

Open the “MMM Example Notebook” in the PyMC-Marketing documentation (pymc-marketing.readthedocs.io). Read it next to tonight's notebook §12 and note the correspondences:

Tonight's notebook	PyMC-Marketing	What to notice
<code>geometric_adstock(x, decay)</code>	<code>GeometricAdstock(l_max=...)</code>	<code>l_max</code> truncates the tail; too small understates slow channels
<code>hill(a, K, s)</code>	<code>LogisticSaturation()</code> (default) or <code>HillSaturation</code>	The logistic has no threshold; Hill has one
<code>DECAY_RANGE</code> (bounded grid)	Priors on <code>alpha</code> (decay) and <code>lam</code> (saturation)	The prior is the bounded grid, made continuous and honest
<code>controls: ramp, season, competitor, pbm_preferred</code>	<code>control_columns=[...]</code> ; <code>yearly_seasonality=...</code>	Same controls; the library can build Fourier seasonality for you
<code>bootstrap + grid sensitivity</code>	posterior draws	One object gives every interval; ROI per draw is a one-liner
<code>geo holdout DiD → compare</code>	lift-test calibration (<code>add_lift_test_measurements</code>)	The experiment becomes part of the likelihood

Reflection B. Jin et al. show that a good fit does not guarantee a good ROI estimate. In tonight's notebook, which channel was the clearest example, and which two facts about its spend pattern explain it? Then: write the prior you would put on that channel's decay parameter, and the one sentence you would use to defend it to a brand team.

C. Promotion at launch — the pharma context (objective 2)

The IQVIA launch framework is in the reading folder from Week 1. Two things it says about promotion matter tonight: the scale of first-year promotional spend, and the fact that the launches it studies won on share of contacts — a dose measure, not a dollar measure.

Despite emerging biopharma's success in product development and commercialization, optimizing first-year sales and first-year promotional spending is a problem. The average first-year sales for top-25 firms was \$114 million versus \$42 million for mid-sized firms and \$28 million for emerging biopharma (see Figure 1). Top-tier firms spent an average of \$53 million on promotional spending during the first year after launch versus \$36 million for midsized firms and \$17 million for emerging biopharma. [...] While top-25 launches return an average of \$61 million, 53.5%, emerging biopharma firms net \$11 million, 39.2%.

— IQVIA, “A Framework for Successful Biopharma Launches” (Gupta & McClellan), Overview — 605 launches, 2007–2016

Takeda had an established sales force of more than 900 reps with broad experience in the same prescriber base through Takeda's diabetes products. The core messages highlighted the limited potential for abuse, proven cardiovascular safety and overall affordability compared to its branded competitors. And Takeda was much more active in promoting Contrave to prescribers than any of its branded competitors. As a result, Contrave displaced all branded competitors in its launch year and became the second largest therapy by active prescribing decisions after generic phentermine.

— IQVIA, “A Framework for Successful Biopharma Launches,” Emphasize the Difference — Contrave (source for the contact data: IQVIA ChannelDynamics)

Note the measurement: IQVIA's own evidence for Contrave's promotion is “share of contacts” from a syndicated promotional-audit panel (ChannelDynamics), not dollars. When you build an MMM for a launch brand, the detailing input is contacts or call-equivalents; spend is the denominator of ROI, not the dose.

High-quality engagement starts with robust customer profiling and flows into strategic orchestration of personal and non-personal channels. MSLs map influence networks, referral routes, formulary constraints,

and evidence needs to prioritise stakeholders with pathway leverage. They then sequence touchpoints to sustain scientific momentum. For example, a focused virtual deep-dive is followed by a compliant, approved email summary and then a peer-to-peer roundtable to progress any evidence gap, with outcomes from each step logged to determine the next best action.

— IQVIA, “HCP Engagement: Medical Affairs” white paper (2025), Insight-driven engagement

Reflection C. The medical-affairs excerpt describes a sequence: virtual deep-dive → approved email → peer roundtable, with outcomes logged. Say why an MMM cannot measure that sequence, which of the three attribution families on slide 6 could, and what the “outcomes logged” data would have to contain.

D. Digital measurement in healthcare — where attribution is closest to solved, and still is not (objective 3)

Two IQVIA papers written for hospital and health-system marketers. They are about paid search and AI search rather than pharma brands, but the measurement problems are exactly the ones a pharma digital channel has — with HIPAA on top.

Move away from ad platform tracking and GA4. Instead, adopt HIPAA-compliant analytics solutions. We have seen clients find success with solutions like Adobe Analytics, Piwik PRO, and custom server-side tracking. [...] Inaccurate data can result in faulty performance reporting, misallocated spend, and missed opportunities for growth.

— IQVIA, “10 Signs Your Agency is Mismanaging Your Paid Search Campaigns” (2025), Tracking and compliance

Search Partner traffic is often of lower quality, sometimes even bot-like, and can artificially inflate metrics like CTR while reducing CPC, making campaigns appear more efficient than they really are. Some agencies leave this setting on by default, which can result in significant budget waste.

— IQVIA, “10 Signs Your Agency is Mismanaging Your Paid Search Campaigns,” Spend-draining campaign defaults

For example, a recent review of a health system’s paid search efforts in support of the Heart and Vascular service line uncovered more than \$75,000 spent on irrelevant keywords and inaccurate geographies over a year-long period. And this was just one service line.

— IQVIA, “10 Signs Your Agency is Mismanaging Your Paid Search Campaigns,” Introduction

Expand beyond search. Reinforce brand trust across a diverse set of channels like CTV, video, and programmatic as traditional search results shrink to accommodate AIOs. Maintain messaging consistency across earned, paid, and owned channels to reinforce signals of trust. Develop a measurement ecosystem to understand impact from impression to appointment and beyond to overcome potential downturns in traditional leading indicators like clickthrough.

— IQVIA, “The State of Search: AI Implications and Actionable Strategies for Healthcare Marketers” (2025), What healthcare marketers should do now

As of May 2025, nearly 90% of healthcare searches on Google return AI-generated summaries. [...] 35% of users switch between multiple search platforms (like Google, ChatGPT, Gemini, and Perplexity) during a single journey. [...] 27% of U.S. adults report using AI instead of traditional search engines.

— IQVIA, “The State of Search,” By the numbers

Reflection D. “Downturns in traditional leading indicators like clickthrough” is a measurement problem: the digital signal the MTA family relied on is disappearing. Explain how that shifts the burden onto MMM and

experiments, and name one thing a pharma brand's digital spend series would need to contain (beyond dollars) for an MMM to measure it.

E. PMSA items for Week 3 (from the syllabus)

- **Axtria/PMSA — “Promotion Response Modeling for Pharma.”** A pharma-specific MMM walkthrough across detailing, samples, conferences, digital and DTC; the real-world counterpart to tonight's synthetic GLUCOREN dataset. insights.axtria.com/blog/promotion-response-modeling
- **Axtria/PMSA — “Marketing-Mix Methodologies to Account for Managed Care Dynamics.”** Extends standard MMM with payer/access overlays — the formulary control from tonight's \$5b, done properly. insights.axtria.com/blog/marketing-mix-methodologies-to-account-for-managed-care-dynamics
- **PMSA 2021 Annual Conference — Axtria-led sessions** including post-COVID HCP engagement adjustments and MMM recalibration. Use as the reference for “what would force you to retrain?”
- **PMSA conference sessions worth bookmarking:** “Dynamic Asymptote Approach to Measure Interaction Among Promotion Channels” (cross-channel synergy, an extension of Hill); “Advanced Targeting for TV and Digital” (calibrating DTC TV and programmatic); “Marketing Mix Modeling and Conferences: Measuring the Impact” (bursty conference spend — the pattern in your homework data). mar-sci.com/marketing-mix-modeling-pharma-conferences/

PMSA member login may be required for some items; the instructor can share slides where licensing permits. The Axtria pair is the recommended reference reading for Homework 1.